

List of unique cities of customers

In [9]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """ select distinct customer_city as CustomerCity from ecommerce.customers"""
cur.execute(query)
data=cur.fetchall()
data
```

Out[9]:

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('senador firmino',),
('viradouro',),
('itabera',),
('cambara',),
('sananduva',),
('tapera',),
('tupaciguara',),
('colombo',),
('ananindeua',),
('coribe',),
('rio doce',),
('mogi mirim',),
('tocos do moji',),
('beberibe',),
('formosa do rio preto',),
('claudio',),
('rio paranaiba',),
('humberto de campos',),
('camocim de sao felix',),
('santo antonio de posse',),
('mesquita',),
('passa quatro',),
('belo oriente',),
('alto araguaia',),
('igarassu',),
('porto velho',),
('cajazeiras',),
('garopaba',),
('rio azul',),
('dois correios',),
('sooretama',),

('cruzeiro do oeste',),
('nazare paulista',),
('riversul',),
('iguaracu',),
('maracas',),
('parana',),
('reboucas',),
('ametista do sul',),
('itu',),
('cacu',),
('vespasiano',),
('lindolfo collor',),
('sao lourenco',),
('santana do sobrado',),
('cafelandia',),
('ibiapina',),
('schroeder',),
('sao joao do oriente',),
('matao',),
('itanhandu',),
('santo angelo',),
('capinopolis',),
('colatina',),
('nova mutum',),
('munhoz de melo',),
('ibate',),
('unai',),
('medeiros neto',),
('anta',),
('ouroeste',),
('central',),
('nova independencia',),
('itapema',),
('sao jose do cedro',),
('camutanga',),
('ipumirim',),
('cristalia',),
('andira',),
('panelas',),
('comodoro',),
('domingos martins',),
('tangua',),
('sao gotardo',),
('araguaina',),
('ilheus',),
('louveira',),
('imbituba',),
('ouricuri',),
('itatinga',),
('santo cristo',),
('arraial d'ajuda',),
('piracanjuba',),
('russas',),
('lambari',),
('vargem alta',),
('sacra familia do tinguá',),
('assis chateaubriand',),
('lago da pedra',),
('unistalda',),

```
('sao joao evangelista',),
('paramirim',),
('goias',),
('paracuru',),
('cruz alta',),
('presidente bernardes',),
('pains',),
('sao goncalo do rio preto',),
('jacinto machado',),
('sobralia',),
('maraba',),
('paraibuna',),
('toropi',),
('jauru',),
('cristalandia',),
('castelo do piaui',),
('ibipora',),
('presidente dutra',),
('vera cruz',),
('paulo frontin',),
...]
```

2. Count the number of orders placed in 2017.

In [10]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """ select count(order_id)  from orders where year(order_purchase_timestamp)=2017
cur.execute(query)
data=cur.fetchall()
data[0][0]
```

Out[10]:

270606

3. Find the total sales per category.

In [10]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
```

```

                                database="ecommerce")
cur = db.cursor()

query= """ select products.product_category category,round(sum(payments.payment_value),2
on products.product_id=order_items.product_id join payments on payments.order_id=order_i
group by category"""
cur.execute(query)
data=cur.fetchall()
df=pd.DataFrame(data)
print(df)

```

		0	1
0		perfumery	7.601080e+07
1	Furniture Decoration		2.145265e+08
2		telephony	7.303231e+07
3	bed table bath		2.568831e+08
4		automotive	1.278441e+08
..		...	...
69	cds music dvds		1.799145e+05
70	La Cuisine		4.370295e+05
71	Fashion Children's Clothing		1.178505e+05
72	PC Gamer		3.261645e+05
73	insurance and services		4.867650e+04

[74 rows x 2 columns]

5.Count of customers from each state

In [13]:

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """select customer_state,count(customer_id) from customers group by customer_stat
cur.execute(query)
data=cur.fetchall()

df=pd.DataFrame(data)
print(df)

```

	0	1
0	RR	276
1	AP	408
2	AC	486
3	AM	888
4	RO	1518
5	TO	1680
6	SE	2100
7	AL	2478
8	RN	2910
9	PI	2970

```
10 PB 3216
11 MS 4290
12 MA 4482
13 MT 5442
14 PA 5850
15 CE 8016
16 PE 9912
17 GO 12120
18 ES 12198
19 DF 12840
20 BA 20280
21 SC 21822
22 PR 30270
23 RS 32796
24 MG 69810
25 RJ 77112
26 SP 250476
```

INTERMEDIATE

1. Number of orders per month in 2018

In [17]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

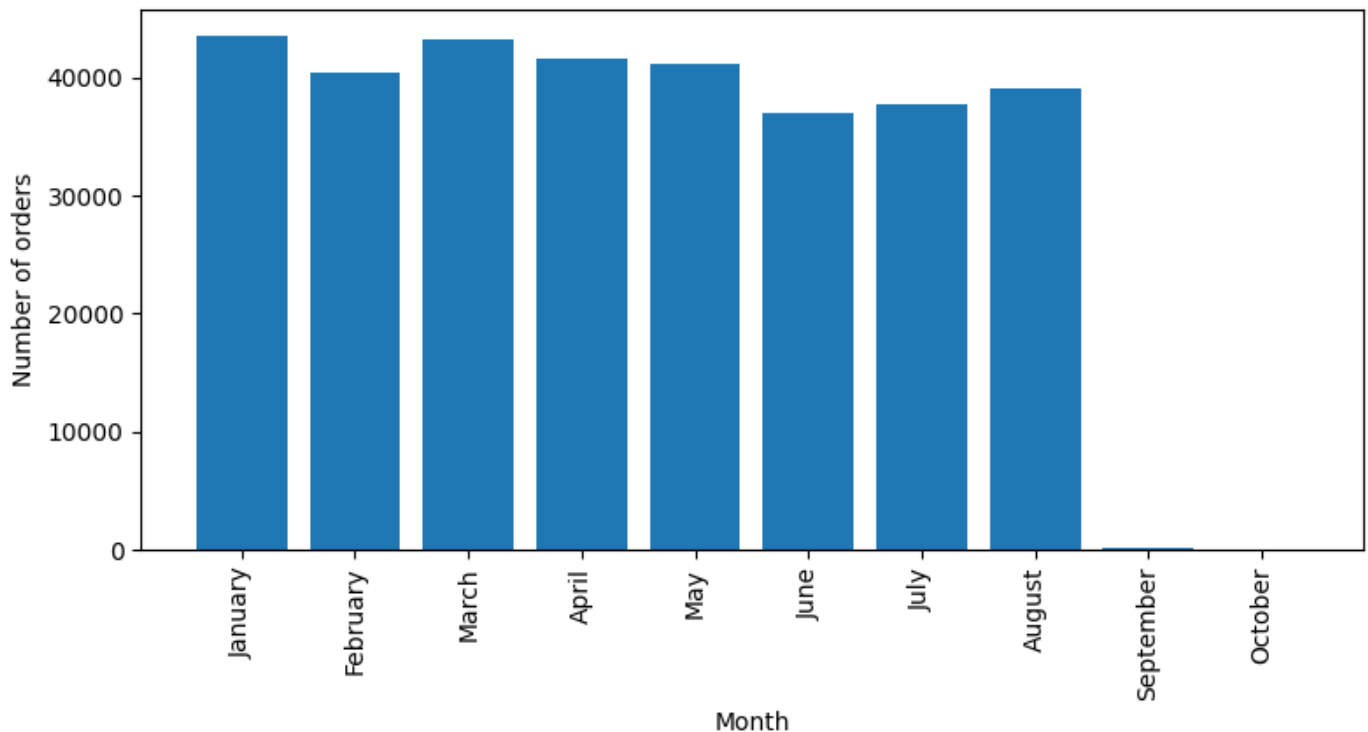
cur = db.cursor()

query= """select monthname(order_purchase_timestamp) month_name,count(*)    Number_of_ord
from orders where year(order_purchase_timestamp)=2018
group by month(order_purchase_timestamp), monthname(order_purchase_timestamp)
order by month(order_purchase_timestamp);"""
cur.execute(query)
data=cur.fetchall()

df=pd.DataFrame(data,columns=["Month","Number of orders"])
print(df)
plt.figure(figsize=(9,4))
plt.bar(df["Month"],df["Number of orders"])
plt.xticks(rotation=90)
plt.xlabel("Month")
plt.ylabel("Number of orders")
plt.show()
```

	Month	Number of orders
0	January	43614
1	February	40368
2	March	43266
3	April	41634

4	May	41238
5	June	37002
6	July	37752
7	August	39072
8	September	96
9	October	24



2. Average number of products per order grouped by customer city

In [11]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """with count_per_order as
(select orders.order_id,orders.customer_id,count(order_items.order_id) as OC
from orders join order_items
on orders.order_id=order_items.order_id
group by orders.order_id,orders.customer_id)
select customers.customer_city, avg(count_per_order.OC) from customers
join count_per_order on count_per_order.customer_id=customers.customer_id
group by customer_city
order by avg(count_per_order.OC) asc;"""
cur.execute(query)
data=cur.fetchall()
```

```
df=pd.DataFrame(data,columns=["customer city","avg product count per order"])
print(df)
```

	customer city	avg product count per order
0	coreau	30.0000
1	alto sao joao	30.0000
2	campo alegre de minas	30.0000
3	manfrinopolis	30.0000
4	ulianopolis	30.0000
...	...	...
4105	matias olimpio	150.0000
4106	candido godoi	180.0000
4107	datas	180.0000
4108	celso ramos	195.0000
4109	padre carvalho	210.0000

[4110 rows x 2 columns]

3.Percentage of total revenue contributed by each product category:

In [16]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """with sales_per_category as (select products.product_category category,round(sum
from products join order_items
on products.product_id=order_items.product_id join payments on payments.order_id=order_i
group by category)
select category,total_sales,(total_sales/sum(total_sales) over())*100 from sales_per_cat
cur.execute(query)
data=cur.fetchall()
df=pd.DataFrame(data)
print(df)
```

	0	1	2
0	perfumery	7.601080e+07	2.495250
1	Furniture Decoration	2.145265e+08	7.042382
2	telephony	7.303231e+07	2.397473
3	bed table bath	2.568831e+08	8.432846
4	automotive	1.278441e+08	4.196812
..	...	...	...
69	cds music dvds	1.799145e+05	0.005906
70	La Cuisine	4.370295e+05	0.014347
71	Fashion Children's Clothing	1.178505e+05	0.003869
72	PC Gamer	3.261645e+05	0.010707
73	insurance and services	4.867650e+04	0.001598

[74 rows x 3 columns]

4. Identify the correlation between the price of the product and the number of times a product has been purchased

In [2]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector
import numpy as np

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """select products.product_category,count(order_items.product_id),
avg(order_items.price) from products join order_items
on products.product_id=order_items.product_id
group by products.product_category; """
cur.execute(query)
data=cur.fetchall()
df=pd.DataFrame(data,columns=[" Product category "," Number_of_times_bought "," Product_price"])
print(df)
arr1=df[" Number_of_times_bought "]
arr2=df[" Product_price "]
a=np.corrcoef([arr1,arr2])
print(a)
```

	Product category	Number_of_times_bought	Product_price
0	HEALTH BEAUTY	290100	130.163531
1	sport leisure	259230	114.344285
2	Cool Stuff	113880	167.357969
3	computer accessories	234810	116.513903
4	Watches present	179730	201.135984
..	...	...	...
69	CITTE AND UPHACK FURNITURE	1140	114.949473
70	Hygiene diapers	1170	40.194615
71	flowers	990	33.637576
72	insurance and services	60	141.644997
73	cds music dvds	420	52.142857

[74 rows x 3 columns]

```
[[ 1.          -0.10631552]
 [-0.10631552  1.          ]]
```

5. Calculate the total revenue by each seller and rank them by revenue

In [7]:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```

import seaborn as sns
import mysql.connector
import numpy as np

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """select sellers.seller_id,count(order_items.seller_id),sum(price+freight_value)
over(order by sum(price+freight_value) desc)
from sellers join order_items
on sellers.seller_id=order_items.seller_id group by sellers.seller_id"""
cur.execute(query)
data=cur.fetchall()
df=pd.DataFrame(data,columns=["Seller_id "," Number of products sold","Revenue_generated"])
print(df)

```

	Seller_id	Number of products sold \
0	4869f7a5dfa277a7dca6462dcf3b52b2	34680
1	7c67e1448b00f6e969d365cea6b010ab	40920
2	53243585a1d6dc2643021fd1853d8905	12300
3	4a3ca9315b744ce9f8e9374361493884	59610
4	fa1c13f2614d7b5c4749cbc52fecda94	17580
...	...	...
3090	702835e4b785b67a084280efca355756	30
3091	4965a7002cca77301c82d3f91b82e1a9	30
3092	1fa2d3def6adfa70e58c276bb64fe5bb	30
3093	77128dec4bec4878c37ab7d6169d6f26	30
3094	cf6f6bc4df3999b9c6440f124fb2f687	30

	Revenue_generated	Seller_rank
0	7.489221e+06	1
1	7.186093e+06	2
2	7.075700e+06	3
3	7.066199e+06	4
4	6.122542e+06	5
...	...	...
3090	5.568000e+02	3091
3091	4.908000e+02	3092
3092	4.770000e+02	3093
3093	4.566000e+02	3094
3094	3.666000e+02	3095

[3095 rows x 4 columns]

Advanced

1.moving average of order_values of each customer over their order history

In [28]:

```

import pandas as pd
import matplotlib.pyplot as plt

```



```

import seaborn as sns
import mysql.connector
import numpy as np

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """SELECT
customer_id,
order_purchase_timestamp,
payment_value,
AVG(payment_value) OVER (
    PARTITION BY customer_id
    ORDER BY order_purchase_timestamp
    ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
) AS mov_avg
FROM (
    SELECT
        o.customer_id,
        o.order_purchase_timestamp,
        SUM(p.payment_value) AS payment_value
    FROM ecommerce.orders o
    JOIN ecommerce.payments p
    ON p.order_id = o.order_id
    GROUP BY o.customer_id, o.order_id, o.order_purchase_timestamp
) AS a;

"""
cur.execute(query)
data=cur.fetchall()
df=pd.DataFrame(data)
print(df)

```

	0	1	2	\
0	00012a2ce6f8dcda20d059ce98491703	2017-11-14 16:08:26	3442.199936	
1	000161a058600d5901f007fab4c27140	2017-07-16 09:40:32	2022.300110	
2	0001fd6190edaaf884bcaf3d49edf079	2017-02-28 11:06:43	5862.599945	
3	0002414f95344307404f0ace7a26f1d5	2017-08-16 13:09:20	5380.500183	
4	000379cdec625522490c315e70c7a9fb	2018-04-02 13:42:17	3210.300064	
...	...	...	...	
99435	fffec9f79fd8c764f843e9951b11341	2018-03-29 16:59:26	2440.800093	
99436	fffed5b6d849fbd39689bb92087f431	2018-05-22 13:36:02	1893.900032	
99437	ffff42319e9b2d713724ae527742af25	2018-06-13 16:57:05	6423.900146	
99438	ffffa3172527f765de70084a7e53aae8	2017-09-02 11:53:32	1365.000000	
99439	ffffe8b65bbe3087b653a978c870db99	2017-09-29 14:07:03	551.100025	
	3			
0	3442.199936			
1	2022.300110			
2	5862.599945			
3	5380.500183			
4	3210.300064			
...	...			
99435	2440.800093			
99436	1893.900032			
99437	6423.900146			

```
99438 1365.000000
99439 551.100025
```

```
[99440 rows x 4 columns]
```

2. Cumulative sales per month for each year

In [18]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector
import numpy as np

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """ select Year,Month,Monthly_payment,sum(Monthly_payment)
over ( partition by Year order by Year,Month) as Cumulative_payments from
(select year(order_purchase_timestamp) as Year,
month(order_purchase_timestamp) as Month,
round(sum(payment_value),2) as Monthly_payment from
ecommerce.orders join ecommerce.payments
on orders.order_id=payments.order_id
group by Year,Month order by Year,Month) as a;
"""

cur.execute(query)
data=cur.fetchall()
df=pd.DataFrame(data,columns=["Year","Month","Monthly_Sales","Cumulative_Sales"])
print(df)
print(" ")

# Year Over year growth :
```

```
data1=df.groupby("Year", as_index=False).tail(1)
print(data1)
data1["Year over year growth"]= data1["Cumulative_Sales"].pct_change() * 100
print(" ")
print(data1)
```

	Year	Month	Monthly_Sales	Cumulative_Sales
0	2016	9	7567.20	7.567200e+03
1	2016	10	1772714.40	1.780282e+06
2	2016	12	588.60	1.780870e+06
3	2017	1	4154641.20	4.154641e+06
4	2017	2	8757240.29	1.291188e+07
5	2017	3	13495907.99	2.640779e+07
6	2017	4	12533640.88	3.894143e+07
7	2017	5	17787564.60	5.672899e+07
8	2017	6	15338291.41	7.206729e+07
9	2017	7	17771487.58	8.983877e+07
10	2017	8	20231889.60	1.100707e+08
11	2017	9	21832873.51	1.319035e+08

12	2017	10	23390336.39	1.552939e+08
13	2017	11	35846483.99	1.911404e+08
14	2017	12	26352044.40	2.174924e+08
15	2018	1	33450125.37	3.345013e+07
16	2018	2	29773900.20	6.322403e+07
17	2018	3	34789563.59	9.801359e+07
18	2018	4	34823564.42	1.328372e+08
19	2018	5	34619464.53	1.674566e+08
20	2018	6	30716415.02	1.981730e+08
21	2018	7	31996222.47	2.301693e+08
22	2018	8	30672759.64	2.608420e+08
23	2018	9	133186.20	2.609752e+08
24	2018	10	17690.10	2.609929e+08

	Year	Month	Monthly_Sales	Cumulative_Sales
2	2016	12	588.6	1.780870e+06
14	2017	12	26352044.4	2.174924e+08
24	2018	10	17690.1	2.609929e+08

	Year	Month	Monthly_Sales	Cumulative_Sales	Year over year growth
2	2016	12	588.6	1.780870e+06	NaN
14	2017	12	26352044.4	2.174924e+08	12112.703758
24	2018	10	17690.1	2.609929e+08	20.000924

C:\Users\User\AppData\Local\Temp\ipykernel_328\2589213023.py:33: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
data1["Year over year growth"] = data1["Cumulative_Sales"].pct_change() * 100
```

3. Calculate the year over year growth:

In [26]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector
import numpy as np

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """with a as
(select year(order_purchase_timestamp) as Year,
round(sum(payment_value),2) as Yearly_payment from
ecommerce.orders join ecommerce.payments
on orders.order_id=payments.order_id
group by Year order by Year)
select Year,Yearly_payment,
(Yearly_payment-lag(Yearly_payment,1) over(order by Year))/(lag(Yearly_payment,1) over(
""")
cur.execute(query)
data=cur.fetchall()
```

```
df=pd.DataFrame(data,columns=["Year","Yearly_payment","YoY_%_Growth"])
print(df)
print(" ")
```

	Year	Yearly_payment	YoY_%_Growth
0	2016	1.780870e+06	NaN
1	2017	2.174924e+08	12112.703759
2	2018	2.609929e+08	20.000924

4. Calculate the retention rate of customers defined as the percentage of customers who make another purchase within 6 months of their first purchase:

In [3]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector
import numpy as np

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """with a as (select customers.customer_id,min(orders.order_purchase_timestamp) a
from
customers join orders
on customers.customer_id=orders.customer_id
group by customer_id),
  b as (select a.customer_id,count(distinct orders.order_purchase_timestamp)
from a join orders
on a.customer_id=orders.customer_id
and orders.order_purchase_timestamp > first_order
and orders.order_purchase_timestamp < date_add(first_order, interval 6 month)
group by a.customer_id)

select 100*( count(distinct b.customer_id))/(count(distinct a.customer_id))
from a left join b
on a.customer_id=b.customer_id
;
"""
cur.execute(query)
data=cur.fetchall()
df = pd.DataFrame(data, columns=["Repeat Customer %"])
print(df)

Repeat Customer %
0          0.0000
```

5. Identify top 3 customers who spent most money every year:

In [27]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector
import numpy as np

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="ecommerce")

cur = db.cursor()

query= """WITH a AS (
    SELECT
        orders.customer_id ,
        YEAR(orders.order_purchase_timestamp) AS Year,
        SUM(payments.payment_value) AS total_money_spent
    FROM ecommerce.orders
    JOIN ecommerce.payments
        ON orders.order_id = payments.order_id
    GROUP BY orders.customer_id, YEAR(orders.order_purchase_timestamp)
),
    b as(SELECT
customer_id,
Year,
total_money_spent,
dense_rank()
over(partition by Year order by total_money_spent desc) as Yearly_ranking
FROM a
)
select *
From b where Yearly_ranking<=3
order by Year,Yearly_ranking
;
"""
cur.execute(query)
data=cur.fetchall()
df = pd.DataFrame(data, columns=["Customer_id","Year","Money_spent","Yearly_Ranking"])
print(df)
```

	Customer_id	Year	Money_spent	Yearly_Ranking
0	a9dc96b027d1252bbac0a9b72d837fc6	2016	42706.501465	1
1	1d34ed25963d5aae4cf3d7f3a4cda173	2016	42022.199707	2
2	4a06381959b6670756de02e07b83815f	2016	36833.400879	3
3	1617b1357756262bfa56ab541c47bc16	2017	409922.402344	1
4	c6e2731c5b391845f6800c97401a43a9	2017	207879.301758	2
5	3fd6777bbce08a352fddd04e4a7cc8f6	2017	201799.804688	3
6	ec5b2ba62e574342386871631fafd3fc	2018	218246.396484	1
7	f48d464a0baaea338cb25f816991ab1f	2018	207666.298828	2
8	e0a2412720e9ea4f26c1ac985f6a7358	2018	144283.198242	3

In []:

